



Professional Summary

Ph.D. candidate in Mathematics and Statistics with an M.S. in Manufacturing Engineering and seven years of combined research, engineering, teaching, and technical management experience. Develops Python and PyTorch systems for semantic and instance segmentation, image denoising, predictive modeling, time-series analysis, reinforcement learning, and robotic perception. Experienced in industrial robotics, Siemens PLC systems, laser-based additive manufacturing, thermal and optical sensing, high-performance computing, and statistically controlled model evaluation. Seeking full-time roles in machine learning, computer vision, robotics, intelligent manufacturing, industrial AI, and applied research.

Technical Skills

Programming and Computing: Python, MATLAB, R, SQL, C, Linux, Git, Jupyter, Papermill, SLURM, HPC, MySQL

Machine Learning and Computer Vision: PyTorch, scikit-learn, OpenCV, NumPy, Pandas, CNNs, ResNet, U-Net, FCN, DeepLabV3, Mask R-CNN, ViT, Swin architectures, Transformers, LSTM, GANs, PPO, transfer learning

Methods and Evaluation: semantic and instance segmentation, object detection, image denoising, time-series forecasting, hyperparameter optimization, class-wise and boundary analysis, IoU/mIoU, Dice, precision, recall, F1, PSNR, SSIM, MS-SSIM, statistical analysis, design of experiments

Robotics and Automation: ABB and KUKA robots, RAPID, RobotStudio, Siemens S7-1500/ET 200SP, TIA Portal, PROFINET, HMI, safety I/O, Raspberry Pi, Arduino, Pixhawk, ArduPilot

Manufacturing and Sensing: laser directed energy deposition, laser powder bed fusion, laser processing, infrared and optical sensing, Fiber Bragg Gratings, galvo scanners, Zaber/Thorlabs stages, G-code generation

Research and Engineering Experience

Graduate Research Assistant

Jan 2021–Present

*University of Texas Rio Grande Valley**Edinburg, TX*

- Develop reproducible Python, PyTorch, and HPC workflows for semantic and instance segmentation, image denoising, robotic perception, reinforcement learning, and manufacturing-process monitoring.
- Design controlled model comparisons using fixed data splits and training protocols, class-wise metrics, boundary measures, computational profiling, learning-curve analysis, and qualitative evaluation.
- Develop infrared-image analysis and data-driven predictive methods for characterizing spatiotemporal thermal evolution in laser directed energy deposition.
- Integrate an ABB robotic LDED platform with Siemens PLC hardware, PROFINET communication, HMI interfaces, safety I/O, RobotStudio workflows, laser systems, powder delivery, and sensing hardware.
- Develop real-time Python control for galvo scanners, lasers, and synchronized Zaber/Thorlabs motion stages and generate process-specific G-code for additive-manufacturing demonstrations.
- Conduct research involving Fiber Bragg Grating sensing in metal additive manufacturing, gaze-tracking cobots for automated metrology, and direct-ink-written multifunctional nanocomposites.
- Prepare technical manuscripts, conference presentations, figures, tables, experimental records, and quantitative reports for interdisciplinary research teams.

Graduate Teaching Assistant

Jan 2022–Dec 2023

*University of Texas Rio Grande Valley**Edinburg, TX*

- Supported Manufacturing Processes Laboratory instruction covering mechanical testing, milling, machining, metallurgy, measurement, and experimental documentation.
- Supported Computer-Aided Design instruction using SolidWorks and guided students through engineering design, modeling, and drawing workflows.
- Mentored student teams in the IDREAM4D internship program on robotics, system integration, troubleshooting, and functional prototype development.

Assistant Manager

Dec 2014–Dec 2020

*Bangladesh Oil, Gas and Mineral Corporation (Petrobangla)**Dhaka, Bangladesh*

- Developed national gas-pricing models for LNG imports and domestic supply and supported production, demand, supply, and financial budgeting.
- Contributed quantitative analysis and briefing materials for four National Gas Budget meetings and national energy-sector planning activities.
- Supported preparation and review of the National Gas Distribution Policy and coordinated technical inputs involving production, transmission, distribution, LNG, and mining entities.
- Prepared technical briefs and analytical responses for parliamentary inquiries, regulators, senior management, and interagency communication.

Selected Machine Learning and Computer Vision Projects

- **Outdoor AMR Semantic Segmentation:** Built a controlled four-class segmentation benchmark comparing U-Net, FCN-ResNet50, and DeepLabV3-ResNet50 for road, sidewalk, cycle-path, and other-region recognition. Developed training, validation, class-wise test evaluation, computational comparison, and qualitative visualization workflows. The work was accepted for oral presentation at SFF Symposium 2026.
- **Instance Segmentation for Automated Inspection:** Prepared multi-class datasets and trained Mask R-CNN and Transformer-based instance-segmentation pipelines. Tuned augmentation and anchor settings and evaluated object localization, masks, class-wise performance, IoU, NMS, and boundary quality.
- **Variational Image Denoising:** Developed and evaluated VAE-based image-denoising methods across Gaussian, salt-and-pepper, and random-uniform noise using repeated-seed testing, Optuna optimization, and SSIM, PSNR, MS-SSIM, MSE, and Charbonnier measures.
- **PPO for Vision-Based Driving:** Developed a CarRacing agent using raw image observations, reward design, checkpointing, and deterministic evaluation; achieved a mean return of 851.39 ± 71.08 over five evaluation episodes.
- **PPO for Continuous Robotic Control:** Implemented BipedalWalker continuous control with reward design, checkpointing, and deterministic evaluation; achieved a mean return of 164.89 ± 3.59 over five evaluation episodes.
- **Explainable Text Classification:** Developed an LSTM sentiment-analysis workflow with tokenization and embeddings and applied LIME and SHAP to identify influential features and improve prediction transparency.
- **Time-Series Forecasting:** Built and compared LSTM and Transformer forecasting models using feature engineering, controlled training, and comparative model evaluation.
- **Generative Image Modeling:** Trained a GAN on celebrity-face images in PyTorch and examined the influence of architecture and training parameters through qualitative evaluation.

Selected Robotics, Automation, and Manufacturing Projects

- **Robotic LDED System Integration:** Integrated an ABB robot, Siemens PLC, PROFINET network, HMI, safety I/O, laser, powder feeder, sensing, and RobotStudio workflows for coordinated deposition experiments.
- **Additive Manufacturing Control and LPBF Demonstration:** Developed real-time Python control for a galvo scanner, laser, and synchronized motion stages and generated G-code from process parameters for coordinated laser-processing demonstrations.
- **Industrial Laser Systems:** Performed calibration and robotic motion programming and operated 1 kW and 8 kW LaserLine systems, including femtosecond-laser coding and laser-manufacturing experiments.
- **Open-Source Pixhawk/ArduPilot Package:** Published `pixhawkcontroller`, a Python package supporting serial, UDP, and TCP communication, telemetry, RC and servo control, and flight-mode switching.
- **ABB RAPID Motion Programming:** Implemented linear, circular, and coordinated industrial-robot motion routines for precise and repeatable automation tasks. Demonstration.
- **Gesture-Controlled Robotic System:** Developed a gesture-based human-robot control interface demonstrated at the UTRGV Defense Manufacturing Expo 2022.
- **Engineering Modeling and Data Systems:** Developed URDF models, OpenAI Gym experiments, Minitab design-of-experiments analyses, SIMIO simulations, CAD designs, and web-scraping pipelines with MySQL.

Education

Ph.D., Mathematics and Statistics with Interdisciplinary Applications, University of Texas Rio Grande Valley Expected Dec 2027
 GPA: 3.93/4.00; research focus: machine learning, computer vision, thermal imaging, robotics, and advanced manufacturing

B.S., Electrical and Electronic Engineering, Chittagong University of Engineering and Technology 2012

Selected Graduate Coursework: Advanced Machine Learning, Deep Learning, Advanced Data Science, Pattern Recognition and Time-Series Data, Advanced Statistical Methods, Advanced Probability and Statistics, Advanced Numerical Analysis, Advanced Biostatistics, Design of Experiments, Simulation of Manufacturing Systems, Engineering Management

Peer-Reviewed Publications and Conference Work

Journal Articles

- Bin Abdus Salam, M. A. R., Ridoy, A. H., Haque, A. K. M. A., Rafique, M. S., Hossain, M. A., **Forhad, M. S.**, et al. "Direct ink writing of shear exfoliated two-dimensional nanomaterial–elastomeric multifunctional nanocomposite." *Materials Today Communications*, vol. 50, 114461, 2026. DOI.
- Ahmed, F., **Forhad, M. S.**, Porag, M. H. "Spectral Behavior of Fiber Bragg Gratings during Embedding in 3D-Printed Metal Tensile Coupons and Cyclic Loading." *Sensors*, vol. 24, no. 12, 3919, 2024. DOI.
- Karunathilake, S., **Forhad, M. S.**, Geng, Z. "Gaze tracking embedded collaborative robots for automated metrology and reverse engineering." *Manufacturing Letters*, vol. 41, pp. 1488–1498, 2024. DOI.

Conference Proceedings and Presentations

- Hallas, K. L., **Forhad, M. S.**, Oraby, T., Peters, B., Li, J. "Comparing life-cycle dynamics of Li-ion batteries clustered by operating conditions with SINDy." *Proceedings of SPIE*, vol. 13036, pp. 31–42, 2024. DOI.
- Islam, A., **Forhad, M. S.**, Li, J., Ahmed, F. "Additive manufacturing of embedded passive electronic components." *Proceedings of the 36th Annual International Solid Freeform Fabrication Symposium*, 2025.
- Messaoudi, S., Duran, K., Garcia, E., **Forhad, M. S.**, et al. "Anomaly detection in UR5 robots using reinforcement learning for smart manufacturing." *Proceedings of the 36th Annual International Solid Freeform Fabrication Symposium*, 2025.
- Meia, M. A., **Forhad, M. S.**, Ramoni, M., Li, J. "Benchmarking U-Net, FCN-ResNet50, and DeepLabV3-ResNet50 for Outdoor AMR Scene Segmentation." Oral presentation, *2026 Annual International Solid Freeform Fabrication Symposium*, Austin, TX, Aug. 2026. Presenting author.
- Forhad, M. S., Meia, M. A., Hallas, K., Rodrigo, H., Ramoni, M., Li, J. "Benchmarking U-Net, FCN-ResNet50, and DeepLabV3-ResNet50 for Outdoor Mobile Robot Scene Segmentation." Submitted for review to IEEE SPICSCON 2026, Track: Computer Vision, Graphics and Virtual Reality, Paper ID 539.

Awards and Recognition

- **Student of the Year Award**, College of Engineering and Computer Science, UTRGV, 2024.
- **IAM Interdisciplinary Excellence Award**, Institute for Advanced Manufacturing, UTRGV, 2024.
- **Outstanding Student Award**, College of Engineering and Computer Science, UTRGV, 2022.
- **Student Appreciation Award**, Institute for Advanced Manufacturing, UTRGV, 2022.
- **Certificate of Appreciation**, IDREAM4, UTRGV, 2021.
- **Presidential Graduate Research Assistantship**, Graduate College, UTRGV, 2020.

Certifications, Training, and Membership

- **Engineer in Training (EIT), Electrical and Power**, Texas, Certificate No. 84003.
- **Google Data Analytics Professional Certificate**; completed two applied capstone projects in R and Python.
- **Nondestructive Testing, Magnetic Particle Testing Level II**, ISO 9712:2005, Bangladesh, 2015.
- Industrial training in electrical engineering and instrumentation, industrial gas utilization efficiency, gas metering and pipeline control, and pipeline design, construction, operation, maintenance, and protection.
- JENESYS 2017, Japan International Cooperation Center, Government of Japan.
- Member, SPIE, the International Society for Optics and Photonics.